

Fake Review Detection and Trust Analytics

CB.SC.U4CSE23213

CB.SC.U4CSE23529

CB.SC.U4CSE23362

CB.SC.U4CSE23661

CB.SC.U4CSE23726

Group 9

19CSE352 Business Analytics

Abstract—Online reviews play a critical role in shaping consumer trust and purchasing decisions in digital marketplaces. However, the growing prevalence of fake and manipulative reviews threatens platform credibility and business integrity. This study focuses on detecting deceptive reviews and analyzing trust dynamics across online review platforms. Using a combination of large-scale review datasets, reviewer metadata, and natural language processing techniques, the project aims to classify fake reviews, assess reviewer credibility, and quantify the impact of review manipulation on consumer trust and business outcomes. By integrating text analytics, behavioral analysis, and trust modeling, the study provides actionable insights for e-commerce platforms, businesses, and policymakers to improve review reliability and enhance user trust.

1 INTRODUCTION

1.1 Business background and motivation

Online reviews have become a critical component of modern digital marketplaces, significantly influencing consumer trust, brand perception, and purchasing decisions. E-commerce platforms, travel and hospitality services, food delivery applications, and app marketplaces rely heavily on user-generated reviews to guide customer choices and rank products or services. Positive reviews often translate directly into higher sales, improved visibility, and stronger brand reputation, making review systems a key business asset.

However, the increasing commercialization of online platforms has led to the widespread emergence of fake and manipulative reviews. Businesses or third-

party agents may post deceptive reviews to artificially inflate ratings, undermine competitors, or mislead potential customers. Such practices distort market fairness, reduce the reliability of recommendation systems, and erode consumer confidence. From a business perspective, the presence of fake reviews can lead to poor customer experiences, higher return rates, loss of long-term trust, and potential regulatory scrutiny.

The motivation for this project arises from the need to preserve trust and transparency in digital ecosystems. Accurate identification of fake reviews enables platforms to protect genuine users, ensure fair competition among sellers, and maintain the credibility of their review systems. Moreover, trust analytics allows businesses to better understand reviewer behavior, assess review credibility, and make data-driven decisions based on reliable feedback.

By applying business analytics, natural language processing, and behavioral analysis techniques, this project aims to support organizations in detecting deceptive reviews and strengthening consumer trust. The outcomes of this study are expected to provide actionable insights for platform administrators, businesses, and policymakers to enhance review authenticity, improve customer satisfaction, and sustain long-term business value.

2 OBJECTIVES

- Enhance consumer trust and review influence

Investigate how online reviews influence consumer trust and purchase decisions, and why some consumers rely heavily on reviews while others remain skeptical, with a focus on review credibility, platform reputation, and user experience.

- Identify characteristics and patterns of fake reviews

Examine which types of reviews are more likely to be fake and which are genuine, and why. Investigate factors such as sentiment extremity, review length, posting frequency, rating behavior, and reviewer history.

- Assess Product-level and platform-level review reliability

Analyze which products or sellers receive a higher proportion of suspicious reviews and which maintain higher trust levels. Investigate the role of category, price range, brand popularity, and platform moderation mechanisms.

- Quantify the impact of fake reviews on business performance

Explore how the presence of fake reviews affects sales rankings, conversion rates, brand perception, and customer retention across online marketplaces.

- Link trust analytics to consumer behavior

Examine whether trust indicators (review credibility scores, verified purchases, reviewer reputation) are associated with higher consumer engagement, repeat purchases, and long-term platform trust.

3 TEAM STRUCTURE AND ROLES

- M.Balaji Sakthivel(Project manager) – Data Acquisition Initial Analysis

Responsible for identifying suitable datasets, understanding data attributes, and performing initial exploratory analysis to understand review distributions and basic patterns.

- M.Hasini Reddy(Data Engineer) – Data Cleaning Feature Preparation

Handles data preprocessing tasks such as cleaning review text, handling missing values, encoding variables, and preparing features required for analysis and modeling.

- Madhav(Data Analyst) – Sentiment Analysis Text Analytics

Focuses on applying sentiment analysis and text-based techniques to extract insights from review content and identify sentiment-related patterns.

- Shivani(Analytics engineer) – Fake Review Identification Trust Metrics

Works on identifying potential fake reviews using analytical or machine learning methods and derives trust-related metrics such as reviewer credibility and review reliability.

- kavin k (business analyst) – Visualization, Interpretation Validation

Responsible for creating visualizations, validating analytical results, interpreting findings from a business perspective.

4 METHODOLOGY

The project adopts a data analytics and machine learning-based methodology to address the trust gap caused by deceptive online reviews. Online review data is collected and subjected to Exploratory Data Analysis (EDA) to examine rating distributions, review frequency, sentiment extremity, and reviewer behavior patterns that indicate potential manipulation.

Textual features are extracted from review content using sentiment and linguistic analysis, and feature dimensionality is reduced using Principal Component Analysis (PCA). Association rule mining is applied to uncover recurring patterns linked to fraudulent feedback. Supervised classification techniques are employed to identify fake reviews, while regression-based models are used to quantify reviewer reliability.

Temporal review activity is analyzed using ARIMA to detect abnormal spikes and coordinated review behavior. The outputs are integrated into an Intelligent Trust Framework that produces credibility scores at reviewer and product levels, enabling more reliable purchasing decisions and fair evaluation of sellers.

Tools Used: Python (Pandas, Scikit-learn, NLP libraries), Excel, Tableau.

5 PROJECT PLAN

5.1 Sprints and timeline

The project follows a structured sprint-based approach to systematically address fake review detection and trust analytics. Each sprint focuses on a distinct analytical phase, progressing from data preparation to modeling, explainability, and business impact evaluation.

5.1.1 *Sprint 1: Data acquisition, cleaning, and exploratory analysis*

Description: Collection and preprocessing of online review data followed by exploratory analysis to understand rating patterns, reviewer behavior, and temporal trends.

Objectives: Prepare high-quality, structured review data and identify early indicators of fake or manipulated reviews.

Key activities: Collect multi-platform online review datasets; clean and pre-

process review text and metadata; perform exploratory data analysis on ratings, review frequency, reviewer activity, and time-based patterns; conduct initial platform-wise and category-wise observations.

Deliverables: Cleaned and preprocessed dataset; exploratory data analysis report with visual insights.

5.1.2 *Sprint 2: Integrated feature engineering and text analytics*

Description: Extraction and integration of textual NLP features and behavioral features to construct a unified feature space for fake review detection.

Objectives: Capture both linguistic and behavioral signals of review fraud and create a comprehensive feature set for modeling.

Key activities: Extract textual features such as sentiment polarity, word frequency, and linguistic patterns; extract behavioral features including review timing, rating deviation, and reviewer activity levels; integrate textual and behavioral features into a unified dataset; perform feature normalization and validation.

Deliverables: Feature-engineered dataset; feature integration and preprocessing documentation.

5.1.3 *Sprint 3: Hybrid modeling, fake review detection, and trust scoring*

Description: Application of hybrid machine learning models to detect fake reviews and compute reviewer credibility and product trust scores.

Objectives: Accurately classify fake and genuine reviews and quantify reviewer trustworthiness beyond binary labels.

Key activities: Train and evaluate traditional machine learning models such as Logistic Regression, Support Vector Machines, and Random Forest; compare performance with advanced models such as LSTM or BERT where feasible; compute reviewer credibility scores and product-level trust indices; perform platform-wise and category-wise vulnerability analysis.

Deliverables: Fake review detection models with evaluation metrics; reviewer trust scores and product trust indices.

5.1.4 Sprint 4: Explainability, impact evaluation, and finalization

Description: Interpretation of model decisions, assessment of business and consumer impact, visualization of insights, and preparation for final presentation.

Objectives: Improve transparency and interpretability of predictions and demonstrate real-world business relevance.

Key activities: Apply explainable AI techniques to interpret model outputs; analyze the impact of fake reviews on ratings, trust, and purchasing decisions; create dashboards and visual analytics; integrate results, document findings, and prepare the end-semester presentation.

Deliverables: Explainability and impact analysis report; final dashboards, project documentation, and presentation slides.

5.2 Agile execution and milestones

The project follows an agile-style execution model, divided into clearly defined sprints with measurable milestones. This approach supports iterative development, early validation of results, and continuous alignment with analytical and business objectives.

5.2.1 Milestones

Milestone 1: Completion of data acquisition, cleaning, and exploratory analysis (end of Sprint 1).

Milestone 2: Completion of integrated feature engineering, hybrid modeling, and trust scoring (end of Sprint 3).

Milestone 3: Completion of explainability analysis, business impact evaluation, and end-semester presentation (end of Sprint 4).

5.3 Sprint overview

Sprint	Focus area
Sprint 1	Data acquisition, cleaning, and exploratory analysis
Sprint 2	Integrated feature engineering and text analytics
Sprint 3	Hybrid modeling, fake review detection, and trust scoring
Sprint 4	Explainability, impact evaluation, and finalization

This agile sprint structure ensures a logical progression from data preparation to model development and impact assessment. The milestone-based approach improves transparency, facilitates review checkpoints, and supports timely and effective project execution.

6 RISK ASSESSMENT

The implementation of a Fake Review Detection and Trust Analytics system involves several technical, data-related, and ethical risks that must be carefully evaluated to ensure reliability and practical applicability.

Data quality and labeling risk Fake review datasets may contain noisy, biased, or incorrectly labeled samples. Inaccurate labels can negatively impact model training and lead to unreliable classification results.

Model bias and generalization risk Models trained on a specific platform or product category may not generalize well to other domains. Linguistic styles and review behavior vary across platforms, increasing the risk of biased predictions.

False positives and false negatives Incorrectly classifying genuine reviews as fake (false positives) can harm user trust, while failing to detect fake reviews (false negatives) may continue to mislead consumers and businesses.

Evolving manipulation techniques Fake review generation methods continuously evolve, especially with the use of AI-generated text. Static models may become outdated if not regularly retrained and updated.

Scalability and performance risk Processing large volumes of reviews in real time may introduce latency and computational challenges, affecting system performance on large-scale platforms.

Ethical and privacy concerns Analyzing user behavior and metadata raises privacy concerns. Improper handling of user data may violate platform policies or data protection regulations.

Business and reputational risk Incorrect or opaque fake review detection decisions may reduce platform credibility and cause dissatisfaction among genuine users or businesses.

7 REFERENCES

REFERENCES

- [1] A. Isiamina, "FakeReviews-RestaurantDataset," GitHub repository, 2024. [Online]. Available: <https://github.com/asiamina/FakeReviews-RestaurantDataset>
- [2] A. Author *et al.*, "Fake review detection using hybrid analytical models," *ScienceDirect*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2772503025000891>
- [3] European Union Open Data Portal, "Fake review-related dataset (Zenodo)," 2023. [Online]. Available: <https://data.europa.eu/data/datasets/oai-zenodo-org-8421611?locale=en>
- [4] D. Dua and C. Graff, "UCI Machine Learning Repository," University of California, Irvine, 2024. [Online]. Available: <https://archive.ics.uci.edu/datasets/?skip=0&take=15&sort=desc&orderBy=NumHits&search=fake>